Formative Assignment 0
The Untyped λ-Calculus

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1. We saw that numbers can be defined in the λ-calculus as follows:

$$
\begin{array}{lcl}
\underline{0} & = & \lambda f.\lambda x.x \\
\underline{1} & = & \lambda f.\lambda x.f\ x \\
\underline{2} & = & \lambda f.\lambda x.f(fx) \\
\underline{3} & = & \lambda f.\lambda x.f(f(fx)) \\
\dots & & \\
\texttt{succ} & = & \lambda a.\lambda f.\lambda x.f\ (a\ f\ x) \\
\texttt{add} & = & \lambda a.\lambda b.\lambda f.\lambda x.a\ f\ (b\ f\ x)
\end{array}
$$

   (a) Prove that ($\texttt{add}\ \underline{2}\ \underline{3}$) β-reduces to $\underline{5}$, i.e., that ($\texttt{add}\ \underline{2}\ \underline{3}$) $\mapsto^*_\beta \underline{5}$. Describe each $\mapsto_\beta$ step.

   (b) Define a multiplication operation `mul` and prove ($\texttt{mul}\ \underline{2}\ \underline{3}$) β-reduces to $\underline{6}$, i.e., that ($\texttt{mul}\ \underline{2}\ \underline{3}$) $\mapsto^*_\beta$ $\underline{6}$. Describe each $\mapsto_\beta$ step.

[4 marks]

2. Instead of using an encoding like in Question 1, we would like an extended λ-calculus with primitive features for numbers. It should have a primitive term `zero`, unary term constructors `succ` and `pred`, and a ternary term constructor `iszero`. The following should be true.

   - The terms `zero`, `succ`(`zero`), `succ`(`succ`(`zero`)), etc. are β-normal forms.
   - `pred`(M) reduces to `zero` if M reduces to `zero`.
   - `pred`(M) reduces to N if M reduces to `succ`(N).
   - `iszero`(M, N, P) reduces to N if M reduces to `zero`.
   - `iszero`(M, N, P) reduces to P if there exists a term Q such that M reduces to `succ`(Q).

   Your tasks are (a) to write down a grammar for the terms of the extended calculus, and (b) to write down the additional rules that need to be added to the definition of β-reduction.

[4 marks]

3. Using the Y combinator, write a function `add'` in the extended calculus of Question 2 that operates on those numbers. Your function should satisfy `add'` $\overline{2}\,\overline{3} \mapsto^*_\beta \overline{5}$ where

$$
\begin{array}{lcl}
\overline{1} & = & \texttt{succ(zero)} \\
\overline{2} & = & \texttt{succ(succ(zero))} \\
\overline{3} & = & \texttt{succ(succ(succ(zero)))} \\
\overline{4} & = & \texttt{succ(succ(succ(succ(zero))))} \\
\overline{5} & = & \texttt{succ(succ(succ(succ(succ(zero)))))}
\end{array}
$$

   You do not need to submit a proof that it does, but we suggest you check to convince yourself.

[2 marks]